

Krishnaa Vadivel | Semiconductor Simulation | Photonics

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Profile

Electrical engineering PhD researcher with broad preparation across thin-film fabrication, semiconductor physics, integrated photonics, device simulation, and scientific software. Combines cleanroom and characterization experience with light-matter modeling, distributed HPC software, and machine learning for semiconductor and photonic systems.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Developed GaN-based power-electronics devices in cleanroom and MOCVD environments during the early PhD, including fabrication-oriented process work, selective-area etching techniques, and materials development.
- Characterized semiconductor materials with SEM, cathodoluminescence, and Raman spectroscopy, connecting fabrication outcomes to optical and structural behavior.
- Built simulation tools for VCSEL emission and coupled photon, phonon, and electron interactions in semiconductor and photonics problems.
- Developed distributed HPC software for exciton-polaron and light-matter interaction modeling using first-principles methods, quantum many-body theory, MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc.
- Used agentic ML and retrieval-based tooling to generate first-principles software inputs from journal literature and technical documentation.
- Maintained strong continuity between theory and engineering practice through coursework and self-developed notes in electromagnetism, signals/systems, semiconductor transport, optics, and device equations.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented related APS work in 2024, 2025, and 2026 on self-trapped excitons and excited-state materials phenomena.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Parallelized CUDA-based software for analysis and visualization of acoustic and nuclear well-tool data used in engineering diagnostics and field interpretation.
- Applied dolfinx-based finite-element analysis to frequency studies of steel pipes in well environments to support physics-driven engineering analysis.
- Contributed instrumentation-oriented engineering around deployed well tools, connecting field data, sensing systems, and numerical analysis.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical articles on lasers, photonics components, optical systems, and engineering applications for a specialized industrial audience.

Selected Projects

Integrated Photonics Band-Pass Filter

- Designed a Meep-based integrated photonics band-pass filter, then fabricated and tested the device with an erbium-doped fiber laser setup.

Semiconductor Device Simulation

- Developed numerical treatments of Poisson, drift-diffusion, and pn-junction-style device equations to connect semiconductor theory with engineering-oriented simulation.

DOLFINx Finite-Element Analysis

- Built 2D DOLFINx/PETSc simulations for transient heat diffusion, Poisson potential around a circular cutout, and incompressible flow past an immersed cylinder, with PyVista and Matplotlib results.

VCSEL Light-Matter Modeling

- Built simulation tools for VCSEL emission and coupled photon, phonon, and electron interactions in semiconductor systems.

Equivariant Neural Networks for Materials and Optics

- Developed PyTorch Geometric equivariant neural networks for material energy prediction and optical-property prediction.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing research spanning computational materials, semiconductor-relevant modeling, and scientific software.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Included work in stochastic computing, semiconductor/device topics, and advanced EE design.

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Included strong preparation in embedded systems, robotics, controls, and hardware-software integration.

Selected Coursework

Semiconductors: Semiconductor Physics; Thin Film Science; drift-diffusion and device modeling foundations

Photonics: Lasers and Optoelectronics; Integrated Photonics

Circuits and VLSI: VLSI Design; Analog VLSI Design; stochastic-computing-related design work

Systems: Mathematics of Signals and Systems; Advanced Embedded Systems

Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell

Fabrication / Characterization: Cleanroom processing, MOCVD, SEM, cathodoluminescence, Raman spectroscopy, laser-bench optical testing

Simulation: Semiconductor device modeling, FDTD photonics simulation, finite-element analysis, first-principles modeling, quantum many-body modeling

Machine Learning: PyTorch, PyTorch Geometric, equivariant graph neural networks, materials- and optical-property prediction, agentic input generation

Scientific Computing: MPI, OpenMP, CUDA, ROCm, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib

Software / Platforms: Python, C, C++, C#, ASP.NET, Microsoft SQL Server, Linux command line, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Frontera, Perlmutter